Ionic Half Equations

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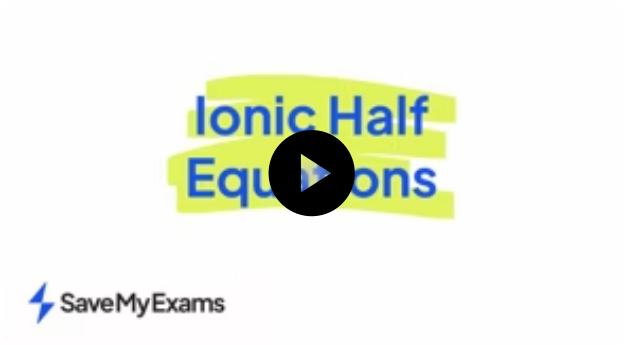

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Ionic half equations

Extended tier only

- Electrochemistry is concerned with the transfer of electrons
  - This is why the definitions of oxidation and reduction are in terms of losing or gaining electrons, not oxygen
  - **Oxidation** is the **loss of electrons**
  - **Reduction** is the **gain of electrons**
- As the ions come into contact with the electrode:
  - Electrons are lost or gained
  - The ions form **neutral** substances
  - These substances are discharged as products at the electrodes
- At the **anode**, negatively charged ions **lose electrons**
  - So, oxidation occurs at the anode
- At the **cathode,** the positively charged ions **gain electrons**
  - So, reduction occurs at the cathode
- **Ionic half-equations** only show **half** of what is happening in a reaction involving electron transfer
  - The ionic half-equation for oxidation shows the loss of electrons
  - The ionic half-equation for reduction shows the gain of electrons

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Writing ionic half-equations

Metals

- Metals are positive ions, e.g. Li^+, Cu^{2+}, Al^{3+}
- If a metal is produced during electrolysis:
  - The metal ions will gain electrons to form metal atoms
  - The metal ions will be reduced

- The charge on the metal ion indicates the number of electrons that will be gained
  - For example:

$$Li^+ + e^- \rightarrow Li$$

$$Cu^{2+} + 2e^- \rightarrow Cu$$

$$Al^{3+} + 3e^- \rightarrow Al$$

Non-metals

- Non-metals are typically negative ions, e.g. Cl^-, Br^-, OH^-
  - The hydrogen ion, H^+, is an exception to this

Hydrogen ion / hydrogen half-equation:

- If hydrogen is formed during electrolysis:
  - **Two** hydrogen ions will **gain** 2 electrons to form hydrogen, H_2
  - Hydrogen ions are **reduced**

$$2H^+ + 2e^- \rightarrow H_2$$

Halide ion / halogen half-equation:

- If a halogen is formed during electrolysis:
  - **Two** halide ions will **lose** 2 electrons to form the halogen
  - Halide ions are **oxidised**

$$2X^- \rightarrow X_2 + 2e^-$$

Hydroxide ion / oxygen half-equation:

- The ionic half-equation for hydroixde ions forming oxygen is more challenging to balance
  - Hydroxide ions **lose** electrons to form oxygen, O_2, and water, H_2O
  - Hydroxide ions are **oxidised**

$$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$$

Table of reduction and oxidation reactions at the electrodes

Electrolysis of	Anode reaction	Cathode reaction
Molten lead(II) bromide, $PbBr_2$	$2Br^- \rightarrow Br_2 + 2e^-$	$Pb^{2+} + 2e^- \rightarrow Pb$
Concentrated aqueous sodium chloride, NaCl	$2Cl^- \rightarrow Cl_2 + 2e^-$	$2H^+ + 2e^- \rightarrow H_2$

Examiner Tips and Tricks

- To help you remember the definitions of oxidation and reduction, use OIL RIG
  - **O**xidation **I**s **L**oss (of electrons) **R**eduction **I**s **G**ain (of electrons)
- To help you remember where oxidation and reduction take place, use a RED CAT and AN OX
  - **RED**uction at the **CAT**hode
  - **AN**ode for **OX**idation

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